Evaluating the informativeness of the Besag-York-Mollié CAR model[☆]

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ABSTRACT

The use of the conditional autoregressive framework proposed by Besag, York, and Mollié (1991; BYM) is ubiquitous in Bayesian disease mapping and spatial epidemiology. While it is understood that Bayesian inference is based on a combination of the information contained in the data and the information contributed by the model, *quantifying* the contribution of the model relative to the information in the data is often non-trivial. Here, we provide a measure of the contribution of the BYM framework by first considering the simple Poisson-gamma setting in which quantifying the prior's contribution is quite clear. We then propose a relationship between gamma and lognormal priors that we then extend to cover the framework proposed by BYM. Following a brief simulation study in which we illustrate the accuracy of our lognormal approximation of the gamma prior, we analyze a dataset comprised of county-level heart disease-related death data across the United States. In addition to demonstrating the potential for the BYM framework to correspond to a highly informative prior specification, we also illustrate the sensitivity of death rate estimates to changes in the informativeness of the BYM framework.

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1. Introduction

The use of spatial models is often motivated by a desire to leverage the spatial structure in the data to improve the precision of estimates from areas with limited data. Due in part to its intuitive and computationally efficient nature, the conditional autoregressive (CAR) model popularized by Besag et al. (1991) (BYM) has become ubiquitous in spatial epidemiology and disease mapping. In addition to being used across a wide range of applications, extensions have been made to spatiotemporal (Waller et al., 1997) and general multivariate (Botella-Rocamora et al., 2015; Gelfand and Vounatsou, 2003) settings. While these and related methods are unquestionably useful, we believe more care should be taken to ensure that these models do not produce estimates that are more precise and more spatially smooth than the data warrant. In particular, missing from the disease mapping literature is a convenient way to quantify the *informativeness* of the BYM framework akin to the concept of “effective sample size” in the Bayesian clinical trials literature (e.g., Morita et al., 2008), perhaps due to the complexity of the conditionally dependent nature of spatial models.

The objective of this paper is simple: to provide guidance for how to measure (or alternatively, *control*) the informativeness of

the BYM framework. Section 2 begins by anchoring our framework in the conjugate Poisson-gamma setting where measuring the informativeness of the prior distribution is trivial. We then propose an approximation of the informativeness of a lognormal prior in Section 2.1 and ultimately the BYM CAR model in Section 2.2. After demonstrating the accuracy of this approximation via simulation in Section 3, we illustrate the potential for oversmoothing using county-level heart disease-related death data in Section 4.

2. Methods

As is the convention when modeling rare event and mortality data (e.g., Besag et al., 1991; Brillinger, 1986), we assume $y_i \sim \text{Pois}(n_i \lambda_i)$, where y_i denotes the number of events in region i from a population of size n_i and λ_i denotes the underlying event rate, for $i = 1, \dots, I$. Since $\lambda_i \sim \text{Gamma}(a_i, b_i)$ is a conjugate prior for the rate parameter, we can write the posterior for λ_i as

$$\lambda_i | y_i, a_i, b_i \sim \text{Gamma}(y_i + a_i, n_i + b_i), \quad (1)$$

which yields the interpretations of a_i and b_i as the “prior number of events” and “prior sample size”, respectively. Furthermore, due to the infinite divisible nature of gamma random variables, we consider y_i , a_i , and $y_i + a_i$ to denote the amount of information contained in the data, the prior distribution, and the posterior distribution, respectively.

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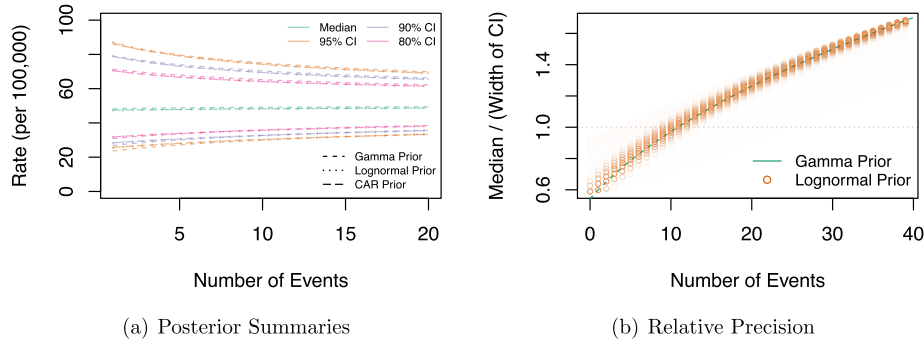


Fig. 1. Posterior comparisons of the gamma and lognormal prior specifications. Panel (a) compares quantiles of the posterior distributions for λ_i for gamma, independent lognormal, and Besag et al. (1991)-inspired CAR priors for various levels of y . Panel (b) compares the relative precision of estimates obtained under the lognormal and gamma prior specifications for various levels y and $E[y|\mu, \sigma^2]$. While the relative precision of an estimate under the gamma prior specification is simply a function of the posterior number of events, the relative precision of an estimate under the lognormal prior specification also appears to be a function of $E[y|\mu, \sigma^2]$; the shading used for each estimate is proportional to $p(y|\mu, \sigma^2)$ to highlight values with high probability of occurring.

2.1. Lognormal approximation

While the prior specification used to construct the posterior in (1) is convenient for illustrating the effect of prior information, it is more common in the disease mapping literature to consider *lognormal* prior specifications for λ_i . Unfortunately, the use of priors like $\lambda_i \sim \text{LogNorm}(\mu_i, \sigma_i^2)$ leads to posterior distributions of an unknown form,

$$p(\lambda_i | y_i, \mu_i, \sigma_i^2) \propto \text{Pois}(y_i | n_i \lambda_i) \times \text{LogNorm}(\mu_i, \sigma_i^2) \\ \propto \exp[-n_i \lambda_i] \times \exp\left[y_i \log \lambda_i - \frac{(\log \lambda_i - \mu_i)^2}{2\sigma_i^2}\right]$$

which can obfuscate the effect of prior information on the posterior distribution. Thus, to better elucidate the effect of prior information when using lognormal priors, we may wish to construct a prior $\lambda_i \sim \text{LogNorm}(\mu_i, \sigma_i^2)$ that contains approximately the same information as $\lambda_i \sim \text{Gamma}(a_i, b_i)$. To achieve this, a natural choice may be to equate the mean and variance of their respective distributions; i.e.,

$$E[\lambda_i | a_i, b_i] = E[\lambda_i | \mu_i, \sigma_i^2] \Rightarrow a_i/b_i = \exp[\mu_i + \sigma_i^2/2] \\ V[\lambda_i | a_i, b_i] = V[\lambda_i | \mu_i, \sigma_i^2] \Rightarrow a_i/b_i^2 = (\exp[\sigma_i^2] - 1) \exp[2\mu_i + \sigma^2]. \quad (2)$$

From (2), we can write $\sigma_i^2 = \log(1/a_i + 1)$ and $\mu_i = \log(a_i/b_i) - \sigma_i^2/2$, which yields $\hat{a}_i = 1/(\exp \sigma_i^2 - 1)$ as an approximation of a_i . Fig. 1 compares quantiles of the posterior distribution for λ given y resulting from a gamma distribution with $a = 8.75$ and y taking values $\{1, 2, \dots, 20\}$ with $a/b = y/n = \lambda_0$ to that resulting from our lognormal approximation, where λ_0 corresponds to a rate of 50 events per 100,000. Moreover, Fig. 1(b) compares the *relative precision* – defined here to be the ratio of an estimate's posterior median to the width of its 95% CI – from the lognormal prior specification to the gamma prior across a range of values of y and $E[y|\mu, \sigma^2]$, where the degree of shading used is proportional to $p(y|\mu, \sigma^2)$ for each level of $E[y|\mu, \sigma^2]$. Here, we see that when $y \approx E[y|\mu, \sigma^2]$ – or rather when observing y is high for a given $E[y|\mu, \sigma^2]$ – the relative precision of the estimates from the lognormal prior specification tend to agree with those from the gamma prior. When an observed y is far from its expected value, however, the relative precision from the lognormal prior specification can deviate from the gamma. As we will see in Section 4, such deviations may be evidence of potential over-smoothing by the model. A deeper discussion of the behavior of the relative precision and additional details regarding the construction of Fig. 1 are provided in Appendix A. Based on these results,

we claim that the prior $\lambda_i \sim \text{LogNorm}(\mu_i, \sigma_i^2)$ is approximately as informative as $\lambda_i \sim \text{Gamma}(a_i, b_i)$ when we define μ_i and σ_i^2 using (2); further support for this claim is provided via simulation in Section 3.

2.2. Extension to CAR model

While (2) yields a measure of the informativeness of independent lognormal prior distributions, spatial models such as the BYM framework utilize *conditionally-dependent* prior distributions. Specifically, if we employ model structures which explicitly model the correlation between λ_i and the remaining λ_j , $j \neq i$, then the informativeness of our model is not dictated by the *marginal* mean and variance of λ_i , but instead by the *conditional* mean and variance, denoted $E[\lambda_i | \lambda_{(i)}, \theta_\lambda]$ and $V[\lambda_i | \lambda_{(i)}, \theta_\lambda]$, respectively, where $\lambda_{(i)}$ denotes the vector $\lambda = (\lambda_1, \dots, \lambda_I)^T$ with the i th element removed and θ_λ denotes a vector of model parameters. In the context of the CAR model of Besag et al. (1991), we begin by assuming

$$\lambda_i | \beta, z, \sigma^2 \sim \text{LogNorm}(\mathbf{x}_i^T \beta + z_i, \sigma^2), \quad (3)$$

where $\mathbf{x}_i$ denotes a $p \times 1$ vector of region-specific covariates with corresponding regression coefficients, β , and z_i denotes a spatial random effect where

$$z_i | \mathbf{z}_{(i)}, \tau^2 \sim \text{Norm}\left(\sum_{j \sim i} z_j / m_i, \tau^2 / m_i\right), \quad (4)$$

where $j \sim i$ denotes that regions i and j are neighbors and m_i denotes the number of regions that neighbor region i . To obtain the necessary conditional distributions for the λ_i , we then integrate $\mathbf{z}$ out of the expression in (3). As shown in Web Appendix B, this leads to a conditional distribution for $\theta_i = \log \lambda_i$ whose precision is bounded below by $1/(\sigma^2 + [\sigma^2 + \tau^2]/m_i)$ – which is achieved for a given m_i when a region neighbors all $I - 1$ of the remaining regions – and bounded above by $1/(\sigma^2 + \tau^2/m_i)$. Here, we use the expression for the lower bound and the approximation in (2) to quantify the model's “informativeness” as

$$\hat{a}_0 = 1/(\exp[\sigma^2 + (\sigma^2 + \tau^2)/m_0] - 1) \quad (5)$$

for a baseline number of neighbors, m_0 . As a general rule of thumb, we will evaluate (5) for $m_0 = 3$ neighbors from this point forward. While (5) will be used to assess and control the model's overall informativeness, county-specific measures, $\hat{a}_i$, can be calculated using the appropriate m_i .

To demonstrate the properties of the Besag et al. (1991)-inspired model from (3) to (4), we considered a scenario consisting of $I = 50$ regions where each region neighbors all of the remaining $I - 1 = 49$ regions (i.e., $m_i = 49$ for all i). We then specified $\sigma^2 = 0.1$ and $\tau^2 = 0.3$, thus constructing a model with $V^{-1}(\theta_i | \beta, \sigma^2, \tau^2, \theta_{(i)}) = 1/(\sigma^2 + [\sigma^2 + \tau^2]/m_i) = 9.25$. Plugging this into the approximation in (2), we obtain $\hat{a}_0 = 8.75$. As illustrated in Fig. 1, this prior specification results in a posterior distribution that is nearly identical to the posteriors resulting from gamma and independent lognormal priors designed to have the same level of information. While this scenario — i.e., $I = 50$ regions that all neighbor each other — is unrealistic, the objective here was simply to illustrate how the expression in (5) can be used to construct priors with the desired properties while avoiding scenarios where spatial models would be inappropriate (e.g., small I).

3. Simulation study

In order to construct the measure of the BYM CAR model's informativeness in (5), we required two steps. First, we needed to establish a relationship between the Poisson-gamma framework and the Poisson-lognormal framework, and then we needed to establish a relationship between an independent lognormal distribution and the BYM CAR model structure. While the latter was mathematically obtained by deriving the conditional distribution for λ_i under the BYM CAR structure in Web Appendix B, we relied on an approximation to establish the relationship the former.

Thus, our goal here is to more thoroughly evaluate the Poisson-lognormal approximation of the Poisson-gamma framework in (2). To this end, we generated data from a Poisson distribution where the underlying rates were sampled from a gamma distribution — $y_i \sim \text{Pois}(n_i \lambda_i)$ and $\lambda_i \sim \text{Gamma}(a, b)$ for $i = 1, \dots, I$, where $a = 5$, $b = a/\lambda_0$, and λ_0 corresponds to a rate of 50 events per 100,000 and where values of n_i are sampled uniformly from 2000 to 40,000 such that $E[y_i | n_i, a, b] \in [1, 20]$. We analyzed these data using both the Poisson-gamma and Poisson-lognormal frameworks with all hyperparameters treated as being unknown with the following prior specifications:

$$\lambda_i \sim \text{Gamma}(a, b), \quad a \sim \text{Unif}(0, 10), \quad \lambda_0 \sim \text{Unif}(0, 10^{-3}) \quad (6)$$

$$\lambda_i \sim \text{LogNorm}(\mu, 1/\gamma), \quad \mu \sim \text{Unif}(-20, 0), \quad \gamma \sim \text{Unif}(0, 10), \quad (7)$$

where $b = a/\lambda_0$ and the bounds on the hyperparameters in (6) and (7) are intended to restrict the parameters to a similar range of values (e.g., when $a \approx 10$, $\gamma \approx 10$). The primary goal of this simulation study will be to assess the degree to which the informativeness of the lognormal prior specification, $\hat{a} = 1/(\exp \sigma^2 - 1)$, is comparable to that of the gamma prior specification, a . As the ability to estimate the hyperparameters in (6) and (7) depends on the amount of data observed, we let $I = \{10, 25, 50, 100, 200\}$; for each level of I , 100 sets of data were generated to better assess the models' performance. All analyses are based on $L = 100,000$ posterior samples obtained using the `rjags` package (Plummer, 2016) with a burn-in of 50,000 iterations and thinned by a factor of 10 to reduce autocorrelation.

Results of the simulation study for $I = 200$ are shown in Fig. 2. In Fig. 2(a), we see that while the informativeness of the gamma prior for $I = 200$ is centered around the true value of $a = 5$, the results from the lognormal prior suggest a slightly less informative posterior. Fig. 2(b) compares the relative precision of the estimates from the two models as function of y , along with the expected curve from the gamma prior. Here, we see a high degree of similarity in the results, albeit with estimates from the lognormal

prior specification tending to have a larger departure from the expected curve than those from the gamma prior specification. This is particularly true when y_i is outside the expected range, $[1, 20]$, reflecting the results displayed in Fig. 1(b). Results for smaller values of I are provided in Web Appendix C. As would be expected, small values of I have much less precision when measuring the informativeness of the priors.

4. Illustrative example: heart disease death data

We now consider a dataset comprised of the number of heart disease-related deaths (ICD-9: 390–398, 402, 404–429) among those aged 35–54 in 1980 from counties in the contiguous (or officially, the *conterminous*) United States. These data predate the CDC's data confidentiality protections — namely, that counts less than 10 are suppressed for data dating back to 1989 (CDC, 2003) — and thus are publicly available without suppression. And while heart disease was the leading cause of death in 1980, mortality rates in this age bracket were still quite low, resulting in a preponderance of small counts and thus motivating the use of spatial models to produce more reliable estimates.

We first consider a case study using the 77 counties of Oklahoma. Here, we will begin by fitting the standard BYM CAR model based on (3) and (4). Vague priors consistent with Bernardinelli et al. (1995) and Waller et al. (1997) will be used — $p(\beta_0) \propto 1$, $\sigma^2 \sim \text{IG}(1, 1/100)$, and $\tau^2 \sim \text{IG}(1, 1/7)$ — and our MCMC algorithm will be run for 100,000 iterations with the first 50,000 iterations discarded as burn-in. After fitting the model, we will estimate the informativeness of this model, $\hat{a}_0$, based on (5). We will then refit the model subject to the restriction that $\hat{a}_0 < 6$ (see Web Appendix D for technical details related to incorporating this restriction in an MCMC algorithm) and explore the implications of this restriction by comparing the two models' estimates and describing how they can lead to different inferential conclusions. Finally, we will repeat these analyses on the remaining 47 states in the contiguous United States — one state at a time — minus those with fewer than five counties where the use of a spatial model may not be appropriate. The goal of this second set of analyses will be to highlight the heterogeneity in the informativeness of the CAR model of Besag et al. (1991) when analyzing the same outcome (heart disease-related deaths) at the same spatial scale (counties) from different locations (states).

4.1. Case study: heart disease-related deaths in Oklahoma

While the heart disease-related death rate for those 35–54 in the state of Oklahoma (106.5 deaths per 100,000) was on par with the national average (108.4), Oklahoma's large number of rural counties led to 64 of the 77 counties experiencing fewer than 10 deaths in this age bracket. This leads to the inferential dilemma which motivates this work — i.e., we want to use models that leverage spatial structure to produce more reliable estimates, but we do not want those models to overwhelm the information contained in the data.

We begin by analyzing the data using the CAR-based model in (3) without restrictions on the informativeness of the prior specification. Initial summaries are shown in Fig. 3. Using the expression for $\hat{a}_0$ in (5), a posterior distribution centered around $\sigma^2 = 0.010$ and $\tau^2 = 0.078$ suggests that the prior specification for this model is estimated to be approximately equivalent to an additional 23 deaths per county, as indicated in Fig. 3(a). The effect of this level of informativeness is reflected in the relative precisions of the rate parameters displayed in Fig. 3(b). Here, we show that the relative precisions of the estimates under the unrestricted CAR model are comparable to those from counties with $23 - 6 = 17$ more ob-

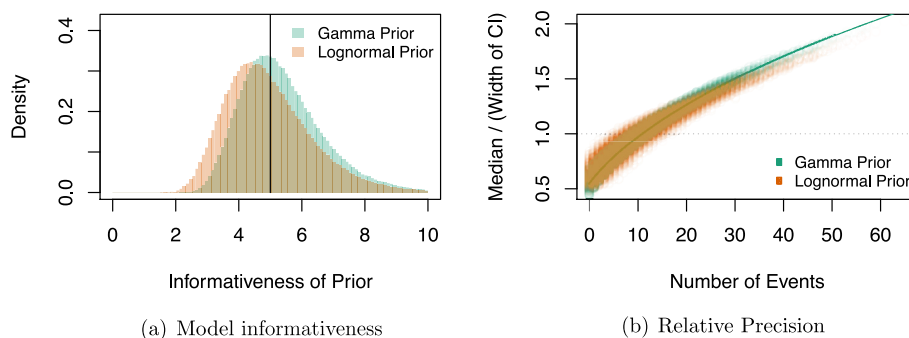


Fig. 2. Posterior summaries from the simulation study for $I = 200$. Panel (a) compares the estimated informativeness of the gamma and lognormal prior specifications relative to the true value of $a = 5$. Panel (b) compares the relative precision of the rate parameters under the two prior specifications relative to the expected curve.

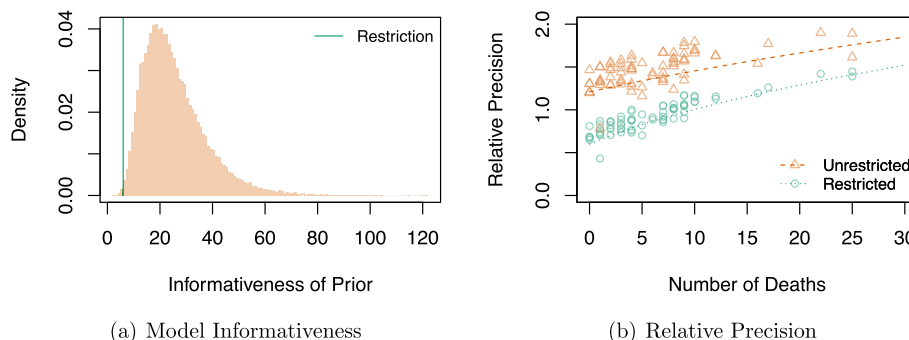


Fig. 3. Posterior summaries from the analysis of heart disease-related death rates in Oklahoma. Panel (a) displays the estimated informativeness of the unrestricted analysis – as measured by $\hat{a}_0$ in (5) – relative to the $\hat{a}_0 < 6$ restriction imposed in the restricted analysis. Panel (b) compares the relative precision of the estimates from the two analyses.

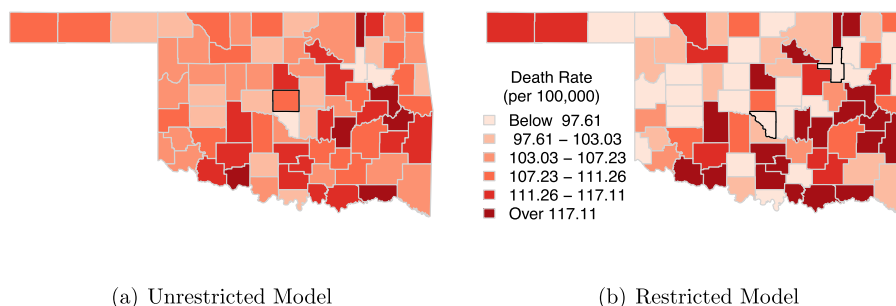


Fig. 4. Comparison of heart disease-related death rates in Oklahoma based on the unrestricted and restricted models. The unrestricted model in Panel (a) contributes approximately 23 deaths per county, compared to just 6 deaths/county in the restricted model in Panel (b). For reference purposes, Oklahoma County is highlighted by a black border in Panel (a) and Cleveland and Tulsa Counties are highlighted in Panel (b).

served deaths than those under a model whose informativeness is restricted to be less than $\hat{a}_0 < 6$.

To see the effect of such strong prior information, we consider the posterior medians of the death rates from the two models' in Fig. 4. Compared to the rates in Fig. 4(b) from the restricted model, the rates in Fig. 4(a) demonstrate the effect of strong spatial smoothing. In particular, many rural counties throughout the state experienced changes of more than 10% toward the statewide average of 106.5 deaths per 100,000, and estimates in much of the central part of the state were elevated due to the effect of an above average rate in Oklahoma County – the state's most populous county. In addition to the rural counties, however, the state's second- (Tulsa) and third-most (Cleveland) populated counties experienced increases of 5% (from 88 deaths per 100,000 to nearly 93) and 11% (from 85 deaths per 100,000 to nearly 95), respectively. We find this is particularly troubling in the case of Tulsa County, as we would have assumed that the 89 deaths observed in that county would be sufficient to guard against the effects of oversmoothing.

Evidence of oversmoothing can also be found by investigating the relative precision of the estimates shown in Fig. 3(b). Specifically, we can use the approach described in Web Appendix A.2 to estimate the informativeness of the model for each estimate by comparing an estimate's relative precision to a noninformative gamma distribution curve. For instance, Fig. 5(a) compares the posterior distribution of the death rates from Muskogee County from the two models, where the data suggest a death rate of 185 deaths per 100,000 based on 25 deaths. Here, we see that the unrestricted model produces a significantly smaller estimate of 130.9 deaths per 100,000. With a relative precision of 1.6, this estimate has an estimated informativeness of 40, indicating that the model is contributing the equivalent of just over $40 - 25 = 15$ additional deaths, far fewer than the 36 additional deaths one would estimate using the posterior medians of σ^2 and τ^2 for a county with $m_i = 6$ neighbors. This appears to be an example of the scenario described in Section 2.1 and shown in Fig. 1(b) where an estimate may have a lower relative precision under the lognormal prior specification than under the analogous gamma prior specification due to

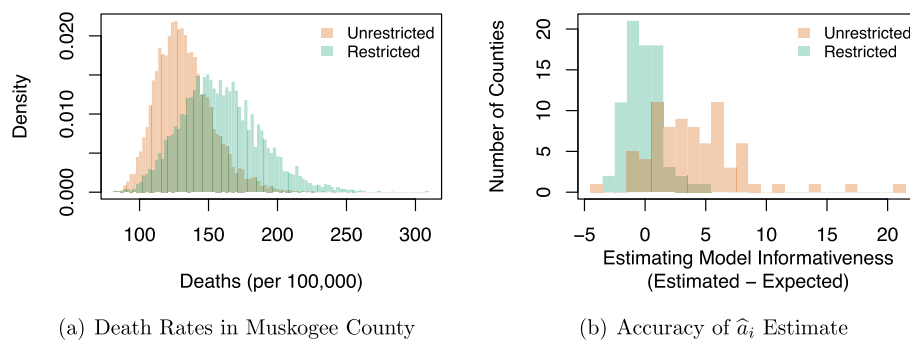


Fig. 5. Evaluation of oversmoothing in heart disease-related death rates in Oklahoma. Panel (a) compares the estimates from the restricted and unrestricted analyses for Muskogee County. Panel (b) compares the county-specific estimates of the model informativeness, $\hat{a}_i$, to estimates based on the relative precision for the two models.

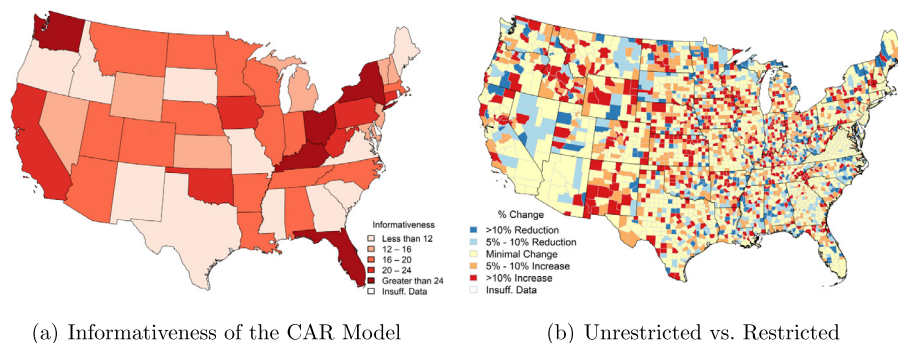


Fig. 6. Results from a state-by-state analysis of the heart disease-related death rates. Panel (a) displays the estimated informativeness of the Besag et al. (1991) model within each state, while Panel (b) displays the percent change resulting from the use of an unrestricted prior specification compared to the restricted (i.e., an "increase" indicates that the unrestricted analysis yields higher rates than one whose informativeness is restricted).

a larger than expected value of y_i . In the case of Muskogee County, this manifests in a posterior distribution that — due to the model's informativeness — is unable to fully break away from the mean of its neighbors. Under the restricted model, however, oversmoothing like that observed in Muskogee County is no longer a significant issue, as the relative precision of the rate estimates more accurately reflect the underlying informativeness of the model, as shown in Fig. 5(b).

4.2. Illustration of heterogeneity in informativeness

We now repeat the above unrestricted analysis on the remaining 47 states in the contiguous United States, minus those with fewer than 5 counties. Fig. 6(a) displays the estimated informativeness measure from (5) for a county with $m_0 = 3$ neighbors from each state. Here, we see that the informativeness of the model in (3) varies wildly, ranging from contributing the effect of $\hat{a}_0 < 5.5$ additional deaths per county in Virginia to $\hat{a}_0 > 36$ in Ohio. While not shown here, there does not appear to be a discernible pattern between our measure of the state-specific measures of model informativeness, $\hat{a}_0$, and their respective event rates or other simple summary statistics (e.g., percent of counties with small counts, percent of rural counties, etc.). Thus, it can be difficult to predict *a priori* the CAR model's informativeness and the extent to which this can alter point estimates (Fig. 6(b)) and their precision.

5. Discussion

The objective of this paper was to provide guidance for how to measure and/or control the informativeness of the BYM CAR model framework. In Section 2, we developed an approximation for the model's informativeness that is a function of the model's spatial and nonspatial variance parameters, τ^2 and σ^2 , respectively. A re-

view of the literature (e.g., Bernardinelli et al., 1995; Waller et al., 1997) suggests the use of relatively noninformative priors for the variance parameters, which may lead users to believe that the BYM framework itself will not be overly informative. As we have illustrated here, this does not appear to be the case.

Furthermore, while much research has been done to construct weakly informative priors (Gelman, 2006) or to theoretically derive prior distributions that penalize complexity (Simpson et al., 2017), the contribution of this paper is to quantify how informative the model is for certain values of σ^2 and τ^2 , regardless of the priors used. Thus, our objective is not to prescribe which priors should be used for these parameters, but instead to provide guidance regarding their interpretation and how restrictions can be imposed on the range of values they are allowed to take. It should also be noted that while this work has focused on the CAR model proposed by Besag et al. (1991), similar methods can (and should) be developed for other popular disease mapping approaches such as the CAR framework of Leroux et al. (2000) and the directed acyclic graph autoregressive model of Datta et al. (2019). While this paper has not derived mathematical expressions to measure the informativeness of these alternative models, the relative precision measure shown in Figs. 1(b) and 3(b) is agnostic to the prior specification used. As such, users of these alternative models can plot the relative precision of their estimates against the true counts and use the approach in Web Appendix A.2 to estimate their model's informativeness.

Regardless of the spatial model used, the implications of overly informative models on inference are numerous. While the most obvious impacts are those that affect our point estimates (e.g., posterior means and medians), perhaps the most substantial impact will be on comparisons between estimates. For instance, such highly informative models may pull disparate rates closer together, obscuring geographic disparities that may be present in the data.

Alternatively, spatial analyses stratified by race may be more likely to indicate significant racial disparities within a county despite data that provide no evidence of such disparities (e.g., in rural, predominantly white counties where such comparisons are merely *hypothetical*). In scenarios like these, we believe a middle path in which spatial structure is incorporated but restricted to a reasonable level would be preferred.

Going forward, we believe there are a few steps that the spatial epidemiology community should consider. First and foremost, users should calculate the informativeness of their models to assess the degree to which the model may be overpowering the information contributed by the data. Ideally, this information would be disclosed in published reports and manuscripts akin to how one's prior distributions are described, thereby prompting users to *defend* their model and its informativeness. Next, experienced users can begin to build constraints into their models to restrict their informativeness — whether that be $\hat{a}_0 < 6$ for a region with $m_0 = 3$ neighbors as considered here or an alternative choice more suitable for their application (e.g., based on the expected number of events per region). This can then lead to developers of R packages and other statistical software to build such mechanisms into their tools, which (as demonstrated in Appendix D) can be achieved with only minor modifications to existing MCMC algorithms. Once this occurs, less statistically savvy users of these models can begin to adopt approaches of this nature in their own research.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.sste.2021.100420](https://doi.org/10.1016/j.sste.2021.100420).

References

- Bernardinelli, L., Clayton, D., Montomoli, C., 1995. Bayesian estimates of disease maps: how important are priors? *Stat. Med.* 14, 2411–2431.
- Besag, J., York, J., Mollié, A., 1991. Bayesian image restoration, with two applications in spatial statistics. *Ann. Inst. Stat. Math.* 43, 1–59.
- Botella-Rocamora, P., Martinez-Beneito, M.A., Banerjee, S., 2015. A unifying modeling framework for highly multivariate disease mapping. *Stat. Med.* 34, 1548–1559.
- Brillinger, D.R., 1986. The natural variability of vital rates and associated statistics. *Biometrics* 42, 693–734.
- CDC, 2003. CDC/ATSDR policy on releasing and sharing data. Manual; Guide CDC-02. Available at <http://www.cdc.gov/maso/Policy/ReleasingData.pdf>. Accessed June 30, 2015.
- Datta, A., Banerjee, S., Hodges, J.S., Gao, L., 2019. Spatial disease mapping using directed acyclic graph auto-regressive (DAGAR) models. *Bayesian Anal.* 14, 1221–1244.
- Gelfand, A.E., Vounatsou, P., 2003. Proper multivariate conditional autoregressive models for spatial data analysis. *Biostatistics* 4, 11–25.
- Gelman, A., 2006. Prior distributions for variance parameters in hierarchical models. *Bayesian Anal.* 1, 515–533.
- Leroux, B.G., Lei, X., Breslow, N., 2000. Estimation of disease rates in small areas: a new mixed model for spatial dependence. In: Halloran, M.E., Berry, D. (Eds.), *Statistical Models in Epidemiology, The Environment, and Clinical Trials*. Springer New York, New York, NY, pp. 179–191.
- Morita, S., Thall, P.F., Müller, P., 2008. Determining the effective sample size of a parametric prior. *Biometrics* 64, 594–602.
- Plummer, M., 2016. rjags: Bayesian graphical models using MCMC. R package version 4–6.
- Simpson, D., Rue, H., Riebler, A., Martins, T.G., Sørbye, S.H., 2017. Penalising model component complexity: a principled, practical approach to constructing priors. *Stat. Sci.* 32, 1–28.
- Waller, L.A., Carlin, B.P., Xia, H., Gelfand, A.E., 1997. Hierarchical spatio-temporal mapping of disease rates. *J. Am. Stat. Assoc.* 92, 607–617.